

Insulin Dose Calculator – Parameter Grid (Ranges)

- **Min/Max (Hard Order Detail Limit):** Unacceptable entry; enforced absolutely during order entry at “add-to-scratchpad.” Out-of-range attempts snap back to the boundary with an alert. The order cannot be signed unless in range, and the calculator shows the invalid indicator: **X**
- **Low Rule/Hi Rule (Soft Order Sign Warning):** Extreme value for age (<1st %ile / >99th %ile); alerted via pop-up warning during order entry at “order sign.” Can be overridden by the prescriber, and the calculator shows the extreme-value indicator for RN awareness and verification: **⚠**
- **Low/Hi Clinical Decision Support (CDS):** The range of values most common for age (the top ~80% of all valid orders for age, generally between the ~10th and 90th %ile). Displayed face-up in context, within the notes section of the PowerPlan during order entry.

Insulin:Carb Ratio (gm Carb:1 Unit insulin)	HARD	RULE	CDS	CDS	RULE	HARD
Age Band	Min	Low Rule	Low CDS	Hi CDS	Hi Rule	Max
0–5 y	1	8	10	50	80	100
6–11 y	1	4	5	30	75	100
12 y +	1	2	4	12	35	100

Correction Factor (Blood Glucose mg/dL:1 Unit insulin)	HARD	RULE	CDS	CDS	RULE	HARD
Age Band	Min	Low Rule	Low CDS	Hi CDS	Hi Rule	Max
0–5 y	4	40	50	200	200	300
6–11 y	4	15	25	100	200	300
12 y +	4	8	15	50	120	300

Target Blood Glucose (mg/dL)	HARD	RULE	CDS	CDS	RULE	HARD
Age Band	Min	Low Rule	Low CDS	Hi CDS	Hi Rule	Max
0–5 y	100	130	150	200	250	250
6–11 y	100	120	130	180	200	250
12 y +	100	120	120	150	200	250

U-10 Dilute Insulin Parameters	HARD	RULE	CDS	CDS	RULE	HARD
	Min	Low Rule	Low CDS	Hi CDS	Hi Rule	Max
Insulin : carb ratio (gm Carb:1 Syringe Unit)	1.5	—	<i>pending</i>	<i>pending</i>	—	40
Correction factor (BG mg/dL:1 Syringe Unit)	10	—	<i>pending</i>	<i>pending</i>	—	200
Target Blood Glucose (mg/dL)	100	130	<i>pending</i>	<i>pending</i>	—	250

Calculator — Entry Guards & Policy	U-100	U-10 Dilute
Carbohydrates eaten	0–300 gm; calculation cap at 250 gm (RN alerted)	0 – 99 gm
Current blood glucose	1–2000 mg/dL; correction cap 600 (RN alerted)	same
Rounding increment	0.5 or 1 Unit (prescriber-set)	1 Syringe Unit
Insulin on board (IOB) Checkbox	suppress correction within 2 h lookback (anti-stacking)	same

Blood-Glucose Decision Thresholds (both concentrations)	Behavior
BG = 0	Critical hypoglycemia — inert; entry minimum is 1, so never triggered
BG < 70	Hypoglycemia — all dosing suppressed; rescue guidance
BG > target	Correction applies (otherwise prandial only)
BG > 600	Correction calculated at 600 cap; prandial unaffected